NMIMS shared use case bank

Thirty five products to build into businesses

Version 3 for 17 and 18 October 2026. Engineering teams build first. MBA teams join active products when they arrive, assess the opportunity, negotiate and recruit, change the product and build its business case. Both groups use the same case ID. Each team has only one active case; engineering cases are unique. The first 20 themes adapt Diwakar's shortlist. C21 to C35 extend the bank with 15 business-oriented products.

A business model in this book is a hypothesis to test. The payer, pricing unit and customer acquisition route are starting points, not a claim that someone will buy. Teams can change the business framing within the selected case, but cannot silently switch to a different product problem. Use the official exchange record if the case changes.

Engineering teams begin with three people: one first-year, one second- or third-year and one final-year. MBA teams have one or two people. Skills listed on each card show the work required; they are not degree, tool or experience prerequisites. A junior engineer can own a real interaction or test, a middle-year engineer a data or business-rule component, and a final-year engineer integration and reliability. These are starting roles, not limits.

All teams receive small synthetic starter fixtures in Starter_Data. They are sufficient to start a bounded workflow, not a full production dataset or an accuracy benchmark. Expand them with labelled synthetic rows and preserve the reference cases. Image fixtures are original illustrations. Live data, live external writes, paid AI access and a cloud deployment are not required. Optional Forge use needs an organiser-provisioned instance and a verified model; a standalone implementation remains valid.

Common engineering outcome

Deliver one complete journey that a judge can repeat, including validation, a useful decision, a review step and a saved result. Provide the runnable product, source, setup instructions, fixture, at least three expected-versus-actual tests, one failure path, known limits and contribution record. Preserve the version shown when MBA partners arrive and demonstrate what changed afterwards.

Common MBA outcome

Choose the customer and payer; explain why this product is the right opportunity; inspect the product and negotiate a meaningful improvement. Build a pricing and unit-economics model, a 30-day growth plan and a six-slide pitch. Record customer evidence or labelled role play, product gaps, negotiations, recruiting offers, outcomes, contribution and the response to a market shock. MBA participants do not have to code, but they must influence a verifiable product or workflow change.

How these cards are scored

Engineering has a common 75-point rubric plus 25 points for the three case checks, weighted 10, 10 and 5 in the order printed. MBA uses a separate 100-point rubric: customer and choice 15; model and economics 20; growth 15; product direction and delivered change 20; negotiation and recruitment 15; pitch 10; contribution and conduct 5. A common four-level evidence scale and the full event rules are in the Programme guide. A useful negotiation or recruitment outcome is judged on evidence and fit, not headcount.

The three engineering checks also give MBA teams a way to challenge the product. A technically passing feature can still be a weak business. Show who benefits, who pays, what costs remain and which experiment would change your mind.

Case index

Case	Product	First payer hypothesis
C01	Review to Action Merchant Copilot	Small D2C quality team
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C20	Slack to Jira Ticket Assistant	Engineering support lead

Case	Product	First payer hypothesis
C21	Order Contribution and Profit Leak Desk	Small D2C founder
C22	Returned Stock Recovery Planner	Returns warehouse operator
C23	AI Pilot Investment Decision Desk	Department head buying AI tools
C24	Creator Campaign Contribution Planner	D2C performance marketing lead
C25	Store Turnaround Decision Simulator	Small retail chain owner
C26	Neighbourhood Delivery Desk	Neighbourhood restaurant owner
C27	Supplier Quote Comparison Desk	Small manufacturer procurement lead
C28	B2B Receivables Follow Up Desk	Small distributor finance lead
C29	Appointment Recovery Desk	Salon or tutoring centre owner
C30	Perishable Stock Recovery Desk	Cafe or grocery owner
C31	Repair Visit Parts Planner	Repair service owner
C32	Group Stay Quote Builder	Small hotel owner
C33	Subscription Renewal Desk	Small SaaS founder
C34	Event Sponsorship Match Desk	Local sponsor or event organiser
C35	Freelancer Scope Change Desk	Small agency owner

C01 Review to Action Merchant Copilot

User and end goal: Merchant quality lead. Choose the product issue to investigate first and the corrective action to test.

Engineering build: Import 20 reviews; show three issue groups, affected products, unique review counts and linked evidence; save a reviewed action.

MBA business assignment: Start with small d2c quality team. Test monthly subscription per store. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Interview three quality leads; test a weekly issue-review trial. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add severity and ownership rather than a word cloud. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Customer research, service operations, value-based pricing. Core.

Starter data: Starter_Data/C01/fixture.json. review_id, product_id, review_text, rating, created_at; a labelled evaluation subset and an action-cost assumption table.

Case outcome checks for engineering:

- 10 points: On 20 judge-labelled reviews, recover at least 16 agreed issue labels; report disagreements and multi-label rules.
- 10 points: Counts reconcile to unique source review IDs; each of the top three issues includes at least two relevant citations.
- 5 points: A duplicate review and an irrelevant instruction do not inflate counts or trigger an action.

Challenge and edge cases: One review mentions two issues; sarcasm; duplicate posts; low-volume severe issue; unknown product.

Path to a real pilot: Shadow a weekly quality review; compare analyst time and reviewed action acceptance against the existing process. Review frequency is not product defect prevalence.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C02 Out of Stock Alternative Finder

User and end goal: Shopper and merchandising manager. Recover a useful shopping option while preserving the shopper's hard requirements.

Engineering build: Choose an unavailable SKU; retrieve up to three available substitutes; explain differences; recalculate after a requirement change.

MBA business assignment: Start with d2c merchandising team. Test monthly fee plus a clearly defined recovery tier. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Run a ten-request substitution concierge test with explicit consent. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add hard requirements and a no-match path. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Merchandising, customer research, contribution analysis. Core.

Starter data: Starter_Data/C02/fixture.json. product_id, category, variant, size, attributes, price, currency, stock, compatibility and timestamp; labelled substitution requests.

Case outcome checks for engineering:

- 10 points: Every result satisfies hard size, compatibility, stock and budget constraints in ten judge scenarios.
- 10 points: At least eight of ten scenarios include a judge-approved substitute, or correctly return none when infeasible.
- 5 points: A stock change or missing mandatory attribute removes the affected recommendation and explains why.

Challenge and edge cases: Exact size missing; only expensive options; stale stock; similar appearance with incompatible specification.

Path to a real pilot: Pilot on one category with human review; measure useful-substitute rate, acceptance and contribution after returns. Do not claim conversion uplift from offline relevance alone.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C03 AI Gift Finder

User and end goal: Shopper and gifting merchandiser. Help a shopper choose a relevant, available gift within a stated budget.

Engineering build: Ask recipient interests, occasion and budget; retrieve up to five catalogue products with grounded reasons; refine after feedback.

MBA business assignment: Start with independent gifting retailer. Test retailer subscription or disclosed referral fee. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test two occasion-specific landing-page propositions. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add corporate bulk gifting or an editable preference flow. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Segmentation, positioning, channel partnerships. Core.

Starter data: Starter_Data/C03/fixture.json. Catalogue IDs, price, availability, attributes and gift-relevant tags; ten labelled personas and explicit exclusions.

Case outcome checks for engineering:

- 10 points: All suggested products exist in the supplied catalogue and respect stock and maximum budget.
- 10 points: At least eight of ten persona tests yield an acceptable candidate under a published relevance guide.
- 5 points: Changing one preference updates the list and explanations without retaining a rejected product.

Challenge and edge cases: Budget below every product; conflicting preferences; too few valid products; no useful recipient detail.

Path to a real pilot: Compare time to shortlist and selection satisfaction with ordinary filters; measure outcomes before introducing personal-history profiling.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C04 Product Image Description Review Tool

User and end goal: Merchant content editor and screen-reader user. Publish accurate, context-appropriate image descriptions with an editor's review.

Engineering build: Process ten supplied images; draft descriptions; edit and approve per image; export product_id and final description.

MBA business assignment: Start with merchant catalogue team. Test usage tiers per reviewed image. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare five manually written descriptions with reviewed drafts. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add reviewer approval and inaccessible-image handling. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Content operations, adoption research, service pricing. Stretch with supplied fallback.

Starter data: Starter_Data/C04/fixture.json. Ten licensed product images, product IDs, surrounding page context and reference descriptions with acceptable variations.

Case outcome checks for engineering:

- 10 points: Ten outputs preserve the image's important purpose and observable details under the reference guide.
- 10 points: No unsupported material, certification, performance or identity claim; uncertainty remains editable.
- 5 points: Review state and CSV export preserve every product ID and the edited text.

Challenge and edge cases: Decorative image; text in an image; multiple products; ambiguous colour; essential information already present next to image.

Path to a real pilot: Have intended users and accessibility reviewers test in page context. This tool alone does not establish site accessibility compliance. W3C guidance is cited in the team review.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C05 Auto Coupon Cart Optimization Agent

User and end goal: Shopper and promotions operator. Find the cheapest permitted payable amount and show why it is valid.

Engineering build: Load a cart and 5-10 offers; enumerate permitted combinations; display eligible/rejected offers, savings, shipping and total.

MBA business assignment: Start with merchant promotions team. Test monthly validation subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Pitch a checkout-error audit to a small retailer. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add margin and shipping transparency to the cheapest-price screen. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Promotion economics, pricing, stakeholder negotiation. Core.

Starter data: Starter_Data/C05/fixture.json. Cart lines, product/category IDs, prices, shipping rule, offer type, threshold basis, cap, exclusions, validity, payment method and stacking policy.

Case outcome checks for engineering:

- 10 points: Match an independent exhaustive reference on ten carts, including every threshold boundary.
- 10 points: Never select an expired, excluded or forbidden stack; final payable reconciles to line amounts and shipping.
- 5 points: A price change recalculates eligibility and invalidates stale selection before confirmation.

Challenge and edge cases: Exactly at minimum spend; post-discount shipping threshold; cap; exclusive versus stackable offer; identical best totals.

Path to a real pilot: Run against a read-only promotions snapshot and compare with checkout's authoritative calculation. Keep final application with the checkout system.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C06 Duplicate Product Review Queue

User and end goal: Catalogue operations lead. Reduce duplicate listings without merging distinct variants.

Engineering build: Inspect 20 products; propose duplicate pairs with matching evidence; accept or reject; export the reviewed queue.

MBA business assignment: Start with marketplace catalogue team. Test monthly review-seat fee. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Offer a manual duplicate audit for one product category. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add review evidence and false-merge cost before bulk automation. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Operations design, quality economics, B2B sales. Core.

Starter data: Starter_Data/C06/fixture.json. product_id, title, brand, model, variant, pack_quantity, unit, image reference; labelled duplicate and hard-negative pairs.

Case outcome checks for engineering:

- 10 points: At least 90% precision on the labelled suggested pairs; show numerator, denominator and recall.
- 10 points: Zero merges of distinct labelled variants; every merge requires explicit reviewer acceptance.
- 5 points: Reviewer decisions persist and rescanning does not duplicate accepted work.

Challenge and edge cases: Two-pack versus single; same title different model; missing brand; transitive clusters with one false pair.

Path to a real pilot: Operate in suggestion mode on one category; measure reviewer time and false-merge rate before enabling any catalogue write.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C07 Budget Based Product Bundle Builder

User and end goal: Shopper assembling a setup. Select a complete, compatible bundle within the total cash budget.

Engineering build: Use one activity and required item slots; choose available items; replace one item; recompute total and validity.

MBA business assignment: Start with work-from-home accessory retailer. Test retailer tool fee or disclosed basket referral fee. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test a budget bundle with five target shoppers. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add compatibility and landed-price comparison. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Category management, packaging, conversion research. Core.

Starter data: Starter_Data/C07/fixture.json. Catalogue IDs, category/slot, price, stock, compatibility attributes, required slots and shipping/tax assumptions.

Case outcome checks for engineering:

- 10 points: Every bundle fills all required slots and satisfies the supplied compatibility matrix.
- 10 points: Total stays within budget on ten reference requests, including delivery charges where defined.
- 5 points: Replacement either preserves validity or explains why no feasible bundle exists.

Challenge and edge cases: Cheapest items incompatible; no stock in one slot; duplicate slot; budget cannot fund the minimum.

Path to a real pilot: Compare usable-bundle completion and returns with ordinary product filters; validate compatibility with category owners.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C08 Natural Language Commerce Analytics Assistant

User and end goal: Merchant operations analyst. Answer a supported business question with reproducible calculations and traceable evidence.

Engineering build: Support five declared question types over supplied data; show answer, date range, chart and query; clarify ambiguity.

MBA business assignment: Start with small retail chain finance lead. Test subscription per store or analyst seat. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Run five real business questions using synthetic data. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add metric definitions and an exportable audit trail. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Financial analysis, metric design, buyer interviews. Core.

Starter data: Starter_Data/C08/fixture.json. Order and return facts with stable IDs, event dates, values, costs and a metric dictionary; ten known-answer questions.

Case outcome checks for engineering:

- 10 points: Exact numeric agreement with ten reference questions, allowing only declared currency rounding.
- 10 points: Queries remain read-only and scoped; every answer displays source, period and metric definition.
- 5 points: Ambiguous period, unsupported metric or missing input triggers clarification instead of fabricated numbers.

Challenge and edge cases: Duplicate joins; empty period; refunds in later month; gross versus net revenue; hostile text in a data field.

Path to a real pilot: Shadow an analyst with a read-only replica; compare correctness, coverage and time while logging unresolved questions.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C09 Screenshot to Shop Assistant

User and end goal: Shopper searching from a photo. Retrieve visually relevant products while showing differences and uncertainty.

Engineering build: Accept one main-product image; return up to five catalogue matches with prices; refine by category or budget.

MBA business assignment: Start with visual-commerce retailer. Test usage-based discovery service. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test ten screenshots and measure useful matches. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add a difference explanation and a manual recovery path. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Search merchandising, customer research, attribution. Stretch with supplied fallback.

Starter data: Starter_Data/C09/fixture.json. Rights-cleared catalogue images and metadata; twenty query images with relevant product labels and hard negatives.

Case outcome checks for engineering:

- 10 points: Report top-5 recall on twenty labelled queries; target at least 16 with a relevant result, with labels visible to judges.
- 10 points: Prices and links resolve to catalogue IDs; hard filters are enforced.
- 5 points: Ambiguous or unrelated input returns uncertainty or no result rather than a claimed exact match.

Challenge and edge cases: Collage; background object; poor crop; similar style different product; missing image.

Path to a real pilot: Pilot on one category; compare top-k relevance and search-to-selection time. Visual similarity does not establish identity or authenticity.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C10 Return Management and Feedback Loop

User and end goal: Returns support reviewer. Make consistent policy-grounded recommendations and surface recurring return reasons.

Engineering build: Enter request and order facts; evaluate supplied policy; explain recommendation; save a local case; aggregate reason counts.

MBA business assignment: Start with retail returns operations team. Test monthly fee tied to case volume. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Map one return workflow and estimate avoidable review effort. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add appeal and exception handling rather than blanket rejection. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Service design, returns economics, policy communication. Core.

Starter data: Starter_Data/C10/fixture.json. Order/item IDs, delivery date, request date, product category, condition and payment status; versioned fictional return policy; reason codes.

Case outcome checks for engineering:

- 10 points: Correct recommendation on ten policy cases including date boundary, excluded item and missing proof.
- 10 points: Every recommendation cites the relevant policy clause and order fact; exceptions route to review.
- 5 points: Repeated submission does not create duplicate cases or inflate reason counts.

Challenge and edge cases: Partial return; holiday timezone boundary; conflicting policy versions; failed case save; missing order.

Path to a real pilot: Begin with reviewer recommendations. Actual refund, stock and payment changes require separate authoritative integrations and approval.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C11 Price Validation and Benchmarking Agent

User and end goal: Pricing operations analyst. Find pricing errors and explain comparable market differences.

Engineering build: Import own and competitor snapshots; match equivalent variants; flag seeded anomalies; show evidence and normalized differences.

MBA business assignment: Start with retail pricing operations team. Test per-store or catalogue subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare ten known pricing mismatches with manual review. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add equivalent-pack matching before price alerts. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Category economics, competitive research, pricing governance. Core.

Starter data: Starter_Data/C11/fixture.json. product_id, brand, model, variant, pack size, units, selling price, MRP, currency, date; competitor observations and matching labels.

Case outcome checks for engineering:

- 10 points: Detect every seeded arithmetic/rule error, including selling price above supplied MRP; identify the triggering rule.
- 10 points: Match at least nine of ten labelled equivalent pairs and reject every incompatible pack/variant in the challenge set.
- 5 points: Missing MRP, stale price or currency mismatch is flagged as unknown rather than compared as fact.

Challenge and edge cases: 2 x 500g versus 1kg; bundle price; currency differences; observed promotion versus regular price.

Path to a real pilot: Use licensed snapshots and category review; measure false alerts and correction turnaround. This exercise does not certify legal pricing compliance.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C12 Damaged Product Evidence Assistant

User and end goal: Support reviewer. Prepare a faithful case summary that separates visible evidence from customer claims.

Engineering build: Upload photos and complaint; summarize visible observations, quoted claims and missing evidence; retain reviewer decision.

MBA business assignment: Start with marketplace support team. Test per reviewed claim or agent-seat fee. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare evidence-review time and escalation accuracy. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add an uncertainty flag and reviewer confirmation. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Support operations, fraud-cost trade-offs, customer fairness. Core.

Starter data: Starter_Data/C12/fixture.json. Rights-cleared images, complaint text, item IDs and annotation guide; benign/ambiguous/damaged examples.

Case outcome checks for engineering:

- 10 points: Correctly distinguish observed details, customer statements and unknowns on ten labelled cases.
- 10 points: No claim about cause, fraud, authenticity or refund eligibility unsupported by the supplied evidence.
- 5 points: Blurry/conflicting photos trigger a precise request for evidence; reviewer edits persist.

Challenge and edge cases: Packaging damage only; old photo; unrelated object; no visible damage despite functional complaint.

Path to a real pilot: Use as reviewer assistance; measure evidence completeness and handling time. Keep approval and fraud decisions with responsible people.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C13 Smart Meeting Preparation Agent

User and end goal: Meeting owner. Arrive with the decisions, unresolved commitments and relevant evidence needed for one meeting.

Engineering build: Use one agenda and 5-10 documents; produce a concise briefing with dated citations and unresolved questions.

MBA business assignment: Start with account management team. Test per-seat monthly subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Prepare three meetings manually and compare usefulness. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add decision history and unresolved issues to the brief. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: B2B sales, account planning, willingness-to-pay research. Core.

Starter data: Starter_Data/C13/fixture.json. Meeting ID/date, agenda, document IDs, authors, timestamps, versions and text; labelled expected open issues.

Case outcome checks for engineering:

- 10 points: Every factual assertion links to the correct source passage and date on the judge pack.
- 10 points: Recover all five seeded open decisions; superseded statements are not presented as current commitments.
- 5 points: Conflicting evidence is disclosed, and unrelated documents stay out of the briefing.

Challenge and edge cases: Cancelled meeting; outdated note; changed owner; conflicting deadlines; source permission boundary.

Path to a real pilot: Pilot with meeting-owner review; measure missing critical issues and preparation time, then approve access scope.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C14 Automated Chaser Draft Queue

User and end goal: Project coordinator. Identify overdue commitments and prepare relevant reminders without chasing completed work.

Engineering build: Extract actions from synthetic conversations; maintain owner/deadline/status; draft reminders; suppress after completion.

MBA business assignment: Start with professional-services delivery manager. Test team subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Observe ten overdue commitments and test actionability. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add stale-message suppression and recipient review. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Project operations, stakeholder communication, adoption. Core.

Starter data: Starter_Data/C14/fixture.json. Message IDs, thread IDs, timestamps/timezone, owners, commitments, due dates and later changes/completions.

Case outcome checks for engineering:

- 10 points: Correct owner, deadline and status for ten labelled commitments, with evidence.
- 10 points: Zero reminders for completed, cancelled or deferred actions in the challenge set.
- 5 points: One action creates one draft; a later deadline update changes the queue without duplicate reminders.

Challenge and edge cases: “Maybe Friday”; timezone; delegated owner; completion in another thread; no response versus not completed.

Path to a real pilot: Shadow a coordinator and measure false reminders, overdue-action coverage and approval rate before enabling controlled sends.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C15 Advanced Daily Summary Agent

User and end goal: Team lead. Understand confirmed progress, blockers and next actions without reading duplicate updates.

Engineering build: Import one day of fictional multi-source updates; deduplicate; produce a cited digest grouped by work item.

MBA business assignment: Start with distributed engineering manager. Test team subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare one daily digest against a manual summary. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add corrections and critical blockers instead of more volume. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Organisational behaviour, communication design, productivity measurement. Core.

Starter data: Starter_Data/C15/fixture.json. Source/message IDs, timestamps, author, work-item ID, text and correction history; five seeded critical facts.

Case outcome checks for engineering:

- 10 points: Recover all five seeded critical facts with correct owner and status, without inventing completion.
- 10 points: Each finding has evidence and duplicate updates count once; later corrections override earlier status visibly.
- 5 points: No access-restricted or unrelated item is included; uncertain status is clearly marked.

Challenge and edge cases: “Almost done”; copied message; overnight timezone; conflicting progress reports.

Path to a real pilot: Compare missed blockers and verification time with existing stand-ups. Keep summaries scoped to the reader’s authorized sources.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C16 Inbox Triage and Reply Draft Assistant

User and end goal: Busy operations professional. Find messages needing action and prepare accurate replies for review.

Engineering build: Process 12 fictional emails; classify urgency/purpose; extract tasks; draft replies; preview label/archive actions.

MBA business assignment: Start with small business operations team. Test per-inbox subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test a narrow invoice/support inbox with synthetic messages. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add escalation and permission controls before send features. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Process design, customer support, SaaS pricing. Core.

Starter data: Starter_Data/C16/fixture.json. Email/thread IDs, sender role, timestamps, subject/body, deadlines and prior replies; labelled action-needed set.

Case outcome checks for engineering:

- 10 points: Recover every seeded critical deadline/action and show precision over the remaining triage decisions.
- 10 points: Drafts reflect the correct thread and make no invented promise, attachment or payment commitment.
- 5 points: A malicious email instruction does not alter permissions or execute actions; user edits and preview work.

Challenge and edge cases: FYI with deadline quoted from old thread; resolved request; ambiguous attachment; spoofed urgency.

Path to a real pilot: Pilot in draft-only mode with narrow permissions; measure missed critical items and draft correction burden.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C17 Voice to Cart Assistant

User and end goal: Shopper using voice. Turn speech into an accurate, reviewable cart proposal.

Engineering build: Record a short request; show editable transcript and parsed quantity; retrieve products; confirm before updating a demo cart.

MBA business assignment: Start with mobile commerce retailer. Test per-store fee or tested assisted-conversion tier. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test five noisy shopping requests with volunteers. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add confirmation, correction and text fallback. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Accessibility research, funnel analysis, category strategy. Stretch with supplied fallback.

Starter data: Starter_Data/C17/fixture.json. Catalogue plus ten consented or synthetic audio requests covering quantities, variants and noisy input; expected cart intents.

Case outcome checks for engineering:

- 10 points: Correct product/quantity intent on at least eight of ten audio requests after the allowed clarification flow.
- 10 points: No cart mutation before explicit confirmation; rejected or edited transcripts update the proposal.
- 5 points: Noisy/ambiguous request asks for clarification and duplicate confirmation cannot double-add.

Challenge and edge cases: “Two” versus “to”; “not red”; multi-item utterance; timeout; spoken injection.

Path to a real pilot: Measure completed cart tasks and correction effort across representative speakers. Test language/accent coverage before rollout.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C18 Checkout Drop Off Investigator

User and end goal: Ecommerce analyst. Find the largest verified funnel deterioration and prioritize an investigation.

Engineering build: Compare two periods; compute session-based step conversion by device/payment segment; show the largest decline with evidence.

MBA business assignment: Start with ecommerce growth lead. Test monthly analytics subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Investigate one observed funnel drop with a known denominator. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add instrumentation gaps and segment comparisons. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Funnel analysis, experimentation, measurement design. Core.

Starter data: Starter_Data/C18/fixture.json. session_id, event_id, event_time, step, device, payment_method, period, success; duplicate/out-of-order events and expected rates.

Case outcome checks for engineering:

- 10 points: Exact agreement on five reference conversion rates using declared session and step rules.
- 10 points: Find the seeded affected segment and absolute percentage-point change without averaging segment rates incorrectly.
- 5 points: Duplicate events and incomplete sessions are handled; correlation is labelled as a hypothesis, not a root cause.

Challenge and edge cases: Device mix shift; payment attempted twice; session crosses midnight; low-volume segment.

Path to a real pilot: Validate event instrumentation first, then compare proposed investigations with analysts. Confirm causality through tests or independent operational evidence.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C19 Inventory Forecast and Replenishment Baseline

User and end goal: Inventory planner. Produce a defensible short-horizon replenishment proposal with uncertainty.

Engineering build: Forecast seven-day demand for 5 SKUs; compare a naive baseline; estimate coverage and proposed quantities with lead times.

MBA business assignment: Start with small chain inventory planner. Test per-location planning subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare a simple baseline forecast and a constrained order plan. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add MOQ and stockout handling before a complex model. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Demand planning, working capital, uncertainty analysis. Stretch with supplied fallback.

Starter data: Starter_Data/C19/fixture.json. date, SKU, units, available/held stock, inbound quantity/ETA, lead time, MOQ, pack size, stockout marker and evaluation cutoff.

Case outcome checks for engineering:

- 10 points: Report per-SKU and aggregate MAE on a later held-out period versus the declared naive baseline; no future data in training.
- 10 points: Replenishment follows the declared stock-position formula, nonnegative quantities, MOQ/pack and inbound timing in five examples.
- 5 points: Zero sales, missing dates and a lead-time shock produce visible uncertainty and a revised proposal.

Challenge and edge cases: New SKU; intermittent demand; inbound late; unavailable stock; forecast below baseline quality.

Path to a real pilot: Use planner shadow review. Better forecasting is not guaranteed; earn credit for honest comparison and sensible decision policy, even if the baseline wins.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C20 Slack to Jira Ticket Assistant

User and end goal: Engineering triage lead. Convert a discussion into one traceable, actionable issue with approval.

Engineering build: Load one fictional Slack thread; preview issue summary, impact and acceptance criteria; check duplicates; confirm creation; retrieve status.

MBA business assignment: Start with engineering support lead. Test per-workspace subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test a ten-message triage workflow and review duplicate rates. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add approval and idempotency before live ticket creation. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Workflow design, internal sales, adoption economics. Core.

Starter data: Starter_Data/C20/fixture.json. Thread/message IDs, timestamps, discussion text; existing issue IDs, titles, status and request idempotency key.

Case outcome checks for engineering:

- 10 points: Issue fields reflect the discussion and missing facts become questions; all acceptance criteria are testable.
- 10 points: Exact/likely duplicates are shown for review and repeated confirmation creates only one issue.
- 5 points: Created issue ID/status is retrievable; timeout/error leaves a truthful recoverable state.

Challenge and edge cases: Conflicting requirements; discussion already ticketed; API timeout after create; changed status.

Path to a real pilot: Pilot with triage-owner approval and limited project scope; measure clarification cycles and duplicates before wider writes.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C21 Order Contribution and Profit Leak Desk

User and end goal: Small brand operator. Locate negative-contribution orders and choose one measurable intervention.

Engineering build: Import order economics; reconcile per-order contribution to channel/product totals; compare a controlled cost/refund scenario; save the decision.

MBA business assignment: Start with small d2c founder. Test monthly contribution-report subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Interview owners about profit leakage and run a sample order audit. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add refund and recovery semantics before a profit headline. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Unit economics, accounting literacy, founder sales. Core.

Starter data: Starter_Data/C21/fixture.json. C02 order economics plus explicit refund_inr, recovered_product_cost_inr and payment_fee_inr; definitions of net revenue and attribution.

Case outcome checks for engineering:

- 10 points: Reference order revenue 1,000 less refund 200, net product cost 400, shipping 50, ads 100 and handling 20 equals INR 230 contribution.
- 10 points: Detail and group totals reconcile; ad allocations and channel spend are never both deducted.
- 5 points: Missing refund facts yield unknown or labelled scenario, and changing return assumptions updates the recommendation.

Challenge and edge cases: Full return with partial recovery; duplicate order; late refund; attribution not causal.

Path to a real pilot: Shadow a finance-approved contribution report. Treat this as operational contribution, not statutory net profit.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C22 Returned Stock Recovery Planner

User and end goal: Warehouse recovery manager. Choose a permitted route for each lot and compare expected net recovery.

Engineering build: Load inspected lots; compare resale/refurbish/scrap scenarios; enforce eligibility/capacity; save a reviewed proposal.

MBA business assignment: Start with returns warehouse operator. Test per-site workflow subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test routing decisions on a small approved stock sample. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add inspection and recovery uncertainty to the routing screen. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Recovery economics, operations, quality control. Core.

Starter data: Starter_Data/C22/fixture.json. C06 lot snapshot plus route eligibility, resale prices, success rates, handling costs and capacities in a clearly labelled assumption table.

Case outcome checks for engineering:

- 10 points: Ten units at resale 100 less refurbishment 20 and handling 5 per unit yield INR 750 future net recovery before other costs; scrap 10 each yields INR 100.
- 10 points: Unsafe/ineligible routes and exceeded capacity are rejected regardless of apparent profit.
- 5 points: Changing inspection confidence or price exposes a scenario, not a guaranteed recovered amount.

Challenge and edge cases: Partial inspection; opened beauty product; unknown condition; handling exceeds recovery.

Path to a real pilot: Pilot on one approved category with quality review; measure realized net recovery and rework rather than forecast revenue.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C23 AI Pilot Investment Decision Desk

User and end goal: Department head. Choose a bounded AI pilot using conservative economics and a measurable stop rule.

Engineering build: Compare 5-8 named workflow pilots; expose cost/benefit assumptions, readiness and overlap; save a portfolio under budget.

MBA business assignment: Start with department head buying ai tools. Test paid pilot review or portfolio subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Run three task-level interviews about time and quality. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add recurring costs, adoption and stop rules. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Capital budgeting, change management, portfolio prioritisation. Core.

Starter data: Starter_Data/C23/fixture.json. C08 portfolio and staff data plus meaningful idea descriptions, annual run cost, adoption, benefit group and pilot budget.

Case outcome checks for engineering:

- 10 points: Pilot costing INR 2 lakh once plus INR 0.6 lakh/year with INR 1.8 lakh/year realized benefit has INR 1.2 lakh annual net benefit and 20-month simple payback, assuming steady benefit from day one.
- 10 points: Portfolio respects budget and prevents double-counting the same hours/revenue across pilots.
- 5 points: Low adoption or failed quality threshold causes an explicit revise/stop decision.

Challenge and edge cases: No recurring cost estimate; overlapping benefit; low data readiness; time freed without cash removal.

Path to a real pilot: Use a small time-boxed pilot with process-owner sign-off. Measure task quality and actual adoption before a wider investment.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C24 Creator Campaign Contribution Planner

User and end goal: Performance marketing lead. Allocate a test budget using contribution scenarios and explicit attribution limits.

Engineering build: Join campaign history and creator roster; show return-adjusted sales and contribution assumptions; propose a capped test allocation.

MBA business assignment: Start with d2c performance marketing lead. Test monthly planning subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare two creator test plans with a capped budget. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add cost and incrementality assumptions to attributed sales. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Growth marketing, attribution, contribution economics. Core.

Starter data: Starter_Data/C24/fixture.json. C20 campaign and roster data plus product margin or unit costs, fulfillment/return costs, incremental attribution assumption and budget.

Case outcome checks for engineering:

- 10 points: 100 orders, 20 returns, INR 1,000 AOV and 10% discount give INR 72,000 retained sales; at 40% pre-fee contribution and INR 20,000 creator fee, scenario contribution is INR 8,800.
- 10 points: Allocations stay within INR 25 lakh; excluded creators remain excluded and overlapping channel attribution is disclosed.
- 5 points: A higher return rate or lower incremental share can reverse the recommendation and updates the screen.

Challenge and edge cases: Unresolved safety review; fee exceeds budget; repeat audience; gross orders mistaken for net orders.

Path to a real pilot: Run a controlled small campaign and measure incremental contribution; do not scale on attributed ROAS alone.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C25 Store Turnaround Decision Simulator

User and end goal: Retail operations head. Choose a testable turnaround, downsize or exit scenario with cash consequences.

Engineering build: Reconcile monthly store profit; compare three six-month scenarios; show break-even assumptions and one-off costs.

MBA business assignment: Start with small retail chain owner. Test per-store planning subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Interview one store manager and compare three turnaround options. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add cash timing and lease exits to a profit-only view. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Retail finance, turnaround planning, stakeholder alignment. Core.

Starter data: Starter_Data/C25/fixture.json. C17 monthly P&L, staffing, category and lease tables; explicit conversion, margin and scenario assumptions.

Case outcome checks for engineering:

- 10 points: INR 10 lakh net sales minus 5 lakh goods, 2 lakh rent, 1 lakh staff and 0.5 lakh other costs gives INR 1.5 lakh operating contribution.
- 10 points: Six-month scenarios include relevant one-off/exit costs and preserve the distinction between profit and cash timing.
- 5 points: Lower conversion or an exit-cost shock updates the recommendation with a clear break-even explanation.

Challenge and edge cases: Online halo unproven; sunk fitout; seasonal month; lease exit timing; sampled staffing data.

Path to a real pilot: Validate cost behavior and lease constraints with store/finance owners; run a bounded trial before committing to closure or expansion.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C26 Neighbourhood Delivery Desk

User and end goal: Small restaurant owner and dispatcher. Accept only orders that can be delivered within the service promise and at a viable contribution.

Engineering build: Build quote, order, rider-capacity check and status updates. Flag late or unprofitable requests and save the reason for a decision.

MBA business assignment: Start with neighbourhood restaurant owner. Test merchant fee plus customer delivery fee. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Pitch one merchant, one customer and one rider role player. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add service radius or a reliable slot before expanding geography. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Marketplace economics, operations, negotiation. Core.

Starter data: Starter_Data/C26/fixture.json. Orders, basket value, merchant fee, customer fee, rider cost, support cost, distance, rider capacity and promised minutes.

Case outcome checks for engineering:

- 10 points: Basket 400 at 15% merchant fee plus customer fee 30, less rider 55, support 8 and payment 5 gives contribution 22.
- 10 points: If rider cost rises to 85, contribution becomes -8; flag it rather than hiding the loss.
- 5 points: If all riders are full, show unavailable capacity and a proposed later slot.

Challenge and edge cases: No rider; a cancelled order; a restaurant that is late; changing costs after a quote.

Path to a real pilot: Test one neighbourhood and a short time window with consent. Measure on-time delivery, repeat use and contribution after refunds.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C27 Supplier Quote Comparison Desk

User and end goal: Procurement manager at a small manufacturer. Compare genuinely equivalent supplier offers and choose a defensible order.

Engineering build: Normalize pack sizes and costs; compare landed cost, lead time and MOQ; save a recommendation with exceptions.

MBA business assignment: Start with small manufacturer procurement lead. Test per-buyer subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Compare three historical quotes with a buyer. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add pack normalization and delivery constraints. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Procurement, supplier negotiation, working capital. Core.

Starter data: Starter_Data/C27/fixture.json. Supplier, item, units per pack, pack price, freight, lead days, MOQ and required delivery date.

Case outcome checks for engineering:

- 10 points: 100 units: supplier A at 10 per unit plus 200 freight totals 1200; B at 11 plus 50 totals 1150.
- 10 points: A five-day promise cannot recommend a seven-day supplier without an explicit exception.
- 5 points: Unknown pack size must block unit-cost comparison.

Challenge and edge cases: Partial fill; tax bases differ; missing pack size; cheap quote with late delivery.

Path to a real pilot: Shadow a buyer for one item family; compare verified savings, lead-time misses and correction effort.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C28 B2B Receivables Follow Up Desk

User and end goal: Finance operations lead at a small distributor. Prioritize overdue invoices and propose appropriate follow-up without treating disputes as routine non-payment.

Engineering build: Import invoices and receipts; calculate overdue balance; separate disputes; draft a reviewable follow-up plan.

MBA business assignment: Start with small distributor finance lead. Test monthly operations subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test a reconciled follow-up list with a finance role player. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add disputes and partial receipts to collection priority. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Receivables operations, cash-flow analysis, communication. Core.

Starter data: Starter_Data/C28/fixture.json. Invoice, customer, due date, as-of date, amount, receipts, dispute status and agreed next action.

Case outcome checks for engineering:

- 10 points: An invoice for 10000 with receipts 4000 has 6000 outstanding.
- 10 points: Paid invoices disappear from collection priority; disputed balances remain visible but enter review.
- 5 points: Changing the as-of date updates ageing consistently.

Challenge and edge cases: Partial payment; unapplied receipt; disputed amount; promise-to-pay already received.

Path to a real pilot: Use read-only records and finance review. Measure reconciliation accuracy and staff effort; do not claim recovered cash from a draft.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C29 Appointment Recovery Desk

User and end goal: Salon or tutoring centre owner. Fill genuinely available appointment slots and reduce missed sessions without overbooking.

Engineering build: Show cancellations and waiting-list matches; record customer consent and rebook; prevent duplicate slot allocation.

MBA business assignment: Start with salon or tutoring centre owner. Test per-location subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test opt-in waitlist messages with five consenting role players. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add service-duration and staff constraints before outreach. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Service operations, customer research, pricing. Core.

Starter data: Starter_Data/C29/fixture.json. Slots, service duration, staff skills, customer availability, opted-in contact flag and booking status.

Case outcome checks for engineering:

- 10 points: A 60-minute request cannot occupy a 30-minute slot.
- 10 points: Two requests competing for one slot result in one confirmed booking and one waitlist entry.
- 5 points: A customer without contact consent is excluded from outreach.

Challenge and edge cases: Late cancellation; staff absence; different service duration; customer withdraws consent.

Path to a real pilot: Pilot one calendar; measure filled cancellations, no-show rate and staff correction effort with consent.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C30 Perishable Stock Recovery Desk

User and end goal: Cafe or small grocery manager. Choose timely markdown or disposal actions while respecting shelf-life and minimum contribution rules.

Engineering build: Show ageing stock; simulate markdowns; allocate quantities and record the recommended action.

MBA business assignment: Start with cafe or grocery owner. Test per-location subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Propose one supervised markdown trial and measure waste. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add expiry checks before an attractive recovery number. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Inventory economics, demand scenarios, operations. Core.

Starter data: Starter_Data/C30/fixture.json. Batch, quantity, expiry time, price, incremental cost, allowed markdown and demand scenarios.

Case outcome checks for engineering:

- 10 points: Ten units sold at 40 with 25 incremental cost each yield 150 contribution.
- 10 points: Expired batches are not sellable even if the margin looks attractive.
- 5 points: Demand above available stock is capped; unsold units stay visible.

Challenge and edge cases: Expiry boundary; cold-chain exception; no demand evidence; damaged stock.

Path to a real pilot: Run a supervised single-store trial with the retailer’s food-safety rules. Track waste, realized recovery and substitutions.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C31 Repair Visit Parts Planner

User and end goal: Service centre dispatcher. Send an appropriately skilled technician with available parts and an honest appointment promise.

Engineering build: Match fault category, technician skill and parts; plan a feasible visit; explain blockers.

MBA business assignment: Start with repair service owner. Test per-technician or location subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Map three missed repair visits and their causes. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add parts availability and diagnostic uncertainty. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Service operations, capacity planning, supplier coordination. Core.

Starter data: Starter_Data/C31/fixture.json. Jobs, fault category, required skill, part, stock, technician availability and visit duration.

Case outcome checks for engineering:

- 10 points: A job needing part P1 cannot be promised when P1 stock is zero.
- 10 points: A technician cannot hold overlapping two-hour visits.
- 5 points: Uncertain diagnosis creates a diagnostic visit, not a guaranteed repair.

Challenge and edge cases: Wrong diagnosis; emergency job; part arrives late; technician unavailable.

Path to a real pilot: Pilot one repair category and measure first-visit completion, repeat visits and promise accuracy.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C32 Group Stay Quote Builder

User and end goal: Small hotel sales manager. Offer a group a feasible package that respects room capacity and a contribution floor.

Engineering build: Build a dated group quote; check room types and nights; calculate package contribution and expiry.

MBA business assignment: Start with small hotel owner. Test per-property subscription or quote workflow fee. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Pitch a group quote to a fictional event organiser. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add room-night availability and quote expiry. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Hospitality sales, yield management, contribution analysis. Core.

Starter data: Starter_Data/C32/fixture.json. Room-night availability, rates, variable cost, group requirements, inclusions and quote expiry.

Case outcome checks for engineering:

- 10 points: Ten rooms for two nights at 2000 per room-night gives 40000 room revenue.
- 10 points: A request for 12 rooms when only 10 are available must offer a partial alternative or decline.
- 5 points: Discounts below the contribution floor require review and cannot auto-confirm.

Challenge and edge cases: Overlapping group holds; breakfast costs; quote expired; partial cancellation.

Path to a real pilot: Shadow the sales desk; compare quote turnaround, acceptance and realized contribution.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C33 Subscription Renewal Desk

User and end goal: Customer-success lead for a small SaaS business. Identify accounts needing a specific renewal action using visible usage and commercial evidence.

Engineering build: Combine renewal dates, activity and service issues; show evidence and a proposed next action; log follow-up.

MBA business assignment: Start with small saas founder. Test per-account-band subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Test a renewal review with five synthetic customer stories. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add usage freshness and a specific owner action. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Customer success, cohort reasoning, SaaS economics. Core.

Starter data: Starter_Data/C33/fixture.json. Account, contract value, renewal date, usage period, active seats, tickets and named owner.

Case outcome checks for engineering:

- 10 points: An account with 2 active seats of 10 shows 20% seat adoption.
- 10 points: An expired reporting window is flagged and cannot be presented as current usage.
- 5 points: A risk label links to evidence and never claims a calibrated churn probability.

Challenge and edge cases: New customer; seasonal usage; missing activity; one severe service issue.

Path to a real pilot: Pilot with a customer-success manager; measure useful actions and retained contribution with an appropriate comparison.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C34 Event Sponsorship Match Desk

User and end goal: Student event organiser and local brand partnerships lead. Build a sponsor package that delivers relevant audience access and credible fulfilment commitments.

Engineering build: Match a sponsor brief to event inventory; price a package; prevent double-selling exclusive slots; track obligations.

MBA business assignment: Start with local sponsor or event organiser. Test paid package-planning service. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Negotiate a sponsor package with an organiser role player. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add exclusivity and fulfilment tracking. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result. MBA: Partnerships, audience research, negotiation. Core.

Starter data: Starter_Data/C34/fixture.json. Sponsor objectives, audience segments, inventory capacity, package price, fulfilment cost and exclusivity.

Case outcome checks for engineering:

- 10 points: Package revenue 30000 less fulfilment cost 18000 gives 12000 contribution.
- 10 points: An exclusive title slot cannot be sold twice.
- 5 points: Unmeasured reach is labelled an estimate, not guaranteed impressions.

Challenge and edge cases: Competing sponsors; prohibited category; cancelled event; overlap in claimed reach.

Path to a real pilot: Use a real organiser review without accepting payments. Measure accepted proposals and completed sponsor deliverables.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.

C35 Freelancer Scope Change Desk

User and end goal: Freelancer or small agency account lead. Turn an ambiguous change request into a clear scope, cost and approval decision.

Engineering build: Compare a request against agreed deliverables; identify changes; estimate effort with assumptions; create a versioned approval record.

MBA business assignment: Start with small agency owner. Test per-seat subscription. Establish a price, cost-to-serve and first-customer path; include the common MBA deliverables.

Growth experiment: Replay three scope requests with a client role player. Record who was approached, the response and what would count as a useful result.

Product change to negotiate: Add approval state and effort capacity before auto-quotes. Save the starting state, negotiated scope and final evidence.

Capabilities to demonstrate: Engineering: UI and state; data validation; tested business rules; CSV or JSON handling; readable evidence and a saved result; optional language/image/speech model with a labelled fallback. MBA: Professional-services economics, client negotiation, scope control. Core.

Starter data: Starter_Data/C35/fixture.json. Original scope, exclusions, request, effort rates, remaining capacity and approved changes.

Case outcome checks for engineering:

- 10 points: An additional six hours at 1000 per hour creates a 6000 change estimate before taxes.
- 10 points: A rejected change does not silently become committed work.
- 5 points: Repeated import of the same request does not double-charge or double-book effort.

Challenge and edge cases: Unclear request; urgent deadline; rework due to own defect; conflicting approvals.

Path to a real pilot: Test with an agency on sanitized examples; measure scope clarity and unapproved effort. It is a workflow aid, not legal advice.

Business proof required: Separate observed evidence from assumptions. Explain the customer value, contribution or willingness-to-pay hypothesis, negotiated change and next 30-day test. No claim of live revenue or production readiness from synthetic data.